

Herald



Herald — pherald Flavor Guide (Operator)

Field	Value
Revision	1
Created	2026-05-30
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Status	active
Status summary	Nano-detail operator reference for pherald (Project Herald) — the richest Herald flavor binary. Documents every subcommand surfaced by <code>pherdald --help</code> (<code>serve</code> , <code>listen</code> , <code>watch</code> , <code>migrate</code> , <code>wizard</code> , <code>commit-push</code> + the §43 GitOps commands), the env/credentials each needs, real example invocations with the binary's actual flags, and which subcommands are live vs not-yet-implemented. ANTI-BLUFF: every claim below was derived from running the built pherald binary (<code>pherdald --help</code> , <code>pherdald <sub> --help</code> , <code>pherdald version --json</code>) and reading <code>pherdald/cmd/pherdald/main.go</code> — nothing invented.
Issues	(none specific to this guide)
Continuation	Bump when <code>migrate down/migrate validate land</code> (currently not-yet-implemented), and when <code>pherdald listen/pherdald watch capture</code> operator-supplied live evidence under <code>docs/qa/</code> .

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§1. What pherald is

pherd is **Project Herald** — flavor key p, prefix PHR, default serving port **24791**. Per `commons/branding.go` its mission line is "Multi-mirror push + submodule propagation + project bindings", and it is the only flavor that carries the full event-ingest path plus the inbound runtime plus the GitOps command catalogue. It is the binary you reach for when a CI job, cron task, or AI agent needs to emit a CloudEvent, drive a Telegram conversation loop, or perform a §2-disciplined commit + multi-mirror push.

Build it (from a Herald checkout with the workspace and submodules initialised):

```
go build -o /tmp/pherd ./pherd/cmd/pherd
```

§2. The subcommand surface

The exact top-level surface, verbatim from `pherd --help`:

Subcommand	What it does	Live?
<code>commit-push</code>	Single-entripoint locked commit + multi-mirror push (§2)	yes
<code>fetch-guard</code>	Pre-edit fetch + rebase enforcement (§11.4.37)	yes
<code>install-upstreams</code>	Configure mirror remotes from <code>upstreams/*.sh</code> declarations (§11.4.36)	yes
<code>listen</code>	Inbound runtime: Telegram <code>getUpdates</code> long-poll + Claude Code dispatch loop	yes
<code>migrate</code>	Apply or inspect <code>commons_storage</code> migrations (spec §9.6)	<code>up/status</code> live; <code>down/validate</code> not-yet-implemented
<code>pre-push</code>	Fetch + investigate + integrate hook (§11.4.71)	yes
<code>reopen</code>	Issues→Fixed reversal + Reopens history (§11.4.55)	yes
<code>serve</code>	Start the Project Herald HTTP server	yes
<code>submodule-propagate</code>	Owned-submodule walk in propagation order (§3)	yes
<code>version</code>	Print Project Herald version + build info	yes
<code>watch</code>	Watch the workable-items SSoT and notify channels on every change	yes
<code>wizard</code>	Interactive credential-setup wizards	yes
<code>completion</code>	Generate shell autocompletion (Cobra built-in)	yes

There are no top-level persistent flags beyond `-h/--help` and `-v/--version`.

§3. version

```
$ pherald version --json
{"arch":"arm64","binary":"pherald","build_time":"unknown","commit":"unknown-dev"}
```

`version` (no flag) prints a human-readable banner; `--json` prints the machine-readable object above. `version`, `commit`, and `build_time` are `-ldflags`-injectable at build time (defaults are `0.0.0-dev / unknown / unknown`). `flavor` is always `p` for this binary.

§4. serve — the HTTP ingest daemon

`pherald serve` is the live HTTP ingest plane. The §32 7-stage Runner (parse → idempotency → tenant → policy → subscriber fan-out → channel dispatch → outcome record) hangs off `POST /v1/events`.

Flags (verbatim from `pherald serve --help`):

Flag	Meaning
<code>--http-port int</code>	TCP+UDP port to bind (default = flavor's <code>DefaultPort</code> , i.e. 24791)
<code>--tls-cert string</code>	Path to PEM-encoded TLS certificate (required when HTTP/3 is enabled, optional for TCP-only)
<code>--tls-key string</code>	Path to PEM-encoded TLS private key (paired with <code>--tls-cert</code>)
<code>--no-brotli</code>	Disable auto-wired Brotli compression middleware (useful for streaming routes that cannot be buffered)

Required environment (the serve plane fails loud at startup without these — by design, per the §107 anti-bluff posture documented at the top of `main.go`):

- `HERALD_PG_DSN` — `postgres://user:pass@host:port/dbname[?sslmode=disable]`. There is **no PG-less serve mode**; the Runner's idempotency archive + subscriber list + evidence writes all require Postgres.
- `HERALD_AUTH_MODE` — `hmac` or `jwtks`. There is **no anonymous serve plane**; `POST /v1/events` is JWT-gated.
- `HERALD_AUTH_HMAC_SECRET` — required when `HERALD_AUTH_MODE=hmac` (32+ random bytes).

Optional environment:

- `HERALD_REDIS_URL` — enables the hot idempotency cache (and the JWKS-mode key cache). The Runner tolerates its absence (PG archive becomes the sole duplicate detector — degraded but functional).
- `HERALD_TGRAM_BOT_TOKEN` — registers the Telegram channel adapter. `null` // is always registered as the sandbox channel.

Example:

```
export HERALD_PG_DSN="postgres://herald:secret@127.0.0.1:24100/herald"
export HERALD_AUTH_MODE=hmac
```

```
export HERALD_AUTH_HMAC_SECRET="$(head -c32 /dev/urandom |
    base64)"
pherald serve --http-port 24791
```

`/v1/healthz`, `/v1/readyz`, and `/metrics` come up immediately and bypass JWT (K8s-probe friendly). `SIGINT/SIGTERM` triggers a graceful shutdown that drains in-flight requests and closes the PG/Redis pools. For the full first-CloudEvent walkthrough see `docs/INTEGRATION.md` §7–§9.

§5. listen — the inbound runtime

`pherald listen` is the long-running inbound loop (Wave 6). It wires `tgram.Subscribe` (Telegram `getUpdates` long-poll) to the production `inbound.Dispatcher`: every inbound message is dispatched through Claude Code (Opus, pinned) per the §32 inbound pipeline, and the parsed `<<<HERALD-REPLY>>>` block is routed by `action(reply / issue.open / event.emit / item mutations)`.

Required environment (verbatim from `pherald listen --help`):

- `HERALD_TGRAM_BOT_TOKEN` — Telegram bot token (validated via `getMe` on boot).
- `HERALD_TGRAM_CHAT_ID` — numeric chat ID; included in the `tgram://` URL.

Optional environment:

- `HERALD_PROJECT_NAME` — Claude Code session name (else `basename(cwd)` else `Herald`).
- `HERALD_CLAUDE_BIN` — path to the `claude` CLI (default: `$PATH` lookup).

Flags:

Flag	Meaning
<code>--qa-out-dir</code> string	Journal every inbound/CC/outbound event to <code><dir>/transcript.jsonl</code> and copy attachments to <code><dir>/attachments/<sha256>.<ext></code> — the Wave 6 T10a §107.x evidence primitive.
<code>--docs-dir</code> string	Docs root containing <code>Issues.md</code> / <code>Fixed.md</code> / <code>Status.md</code> / <code>CONTINUATION.md</code> / <code>Help.md</code> . Default docs. <code>Issues.md</code> MUST exist — startup fails otherwise (Wave 6.5 T7 §107 fail-loud).
<code>--db</code> string	Workable-items SQLite DB path backing <code>item.update/item.delete/confirmed-investigation actions</code> (default <code>\$HERALD_WORKABLE_DB</code> or <code>docs/workable_items.db</code> ; HRD-152). An empty path with no env makes those actions return an explicit "no ItemMutator configured" error.

Signal handling: `SIGINT/SIGTERM` cancels the long-poll cleanly via `signal.NotifyContext`.

```
export HERALD_TGRAM_BOT_TOKEN=80000000000:XXXX
export HERALD_TGRAM_CHAT_ID=987654321
pherald listen --qa-out-dir docs/qa/manual-run-$(date +%s)
```

NOTE: `listen` requires real Telegram credentials and a `claude` CLI to do anything useful end-to-end; without them it fails loud at boot. Fully-autonomous (no-

human) testing of this loop is the qaherald flavor's job — see docs/guides/QAHERALD.md.

§6. watch — workable-items SSoT → notify

pherald watch is a second long-running daemon (HRD-153, ATMOSphere integration). It composes the workable-items single-source-of-truth **watch** → **diff** → **notify** flow:

1. Opens the shared workable-items SQLite DB (--db) and snapshots every item at the watched locations (Issues + Fixed).
2. Starts a commons_watch.Watcher (fsnotify + WAL-poll) on the DB file and the Markdown trackers (--issues / --fixed).
3. On every change: re-lists current items, diffs against the prior snapshot, renders each per-property delta, and fans it out through the production ChannelDispatcher to every configured recipient's channel.

Channel/recipient config reuses the same HERALD_CHANNELS + per-channel namespaced env as pherald listen (the production fan-out path) — see docs/guides/MESSENGER_CHANNELS.md.

Flags:

Flag	Default	Meaning
--db string	\$HERALD_WORKABLE_DB or docs/workable_items.db	Workable-items SQLite DB path
--issues string	docs/Issues.md	Issues.md tracker path (watched)
--fixed string	docs/Fixed.md	Fixed.md tracker path (watched)
--poll duration	1s	WAL-poll fallback interval (0 disables, fsnotify only)

```
pherald watch --db docs/workable_items.db --poll 1s
```

SIGINT/SIGTERM cancels the watch loop cleanly. The watcher mechanics are documented in docs/guides/COMMONS_WATCH.md; the store in docs/guides/COMMONS_WORKABLE.md.

§7. migrate — schema migrations

pherald migrate runs Herald's embedded SQL migrations against Postgres. It requires HERALD_PG_DSN (no silent default).

Subcommand	State	Behaviour
up	live	Apply every pending migration in version order.
status	live	Report the highest applied migration version.
down	not yet implemented	Roll back the most recent migration (destructive op per §9.1, future HRD).
validate	not yet implemented	Audit applied migrations vs the embedded set (schema-drift detection, future HRD).

```
export HERALD_PG_DSN="postgres://herald:secret@127.0.0.1:24100/
    herald"
pherald migrate up
pherald migrate status      # e.g. "schema version: 12"
```

down and validate exist as subcommands but return a not-yet-implemented message — do not script against them yet.

§8. wizard credentials — credential setup

pherald wizard credentials [service] is the canonical credential-setup tool. Supported services (LIVE): telegram (HRD-011 — bot token + chat ID, validated via getMe + getChat), claude-code (HRD-012 — claude binary + project name, validated via --version), and all (both, in order). Without an argument it prompts from a menu (interactive only).

Resolution order is **flag** → **env** → **prompt**. Flags:

Flag / env	Service
--bot-token / HERALD_TGRAM_BOT_TOKEN	Telegram
--chat-id / HERALD_TGRAM_CHAT_ID	Telegram (skips getUpdates polling)
--claude-bin / HERALD_CLAUDE_BIN	Claude Code
--claude-project / HERALD_CLAUDE_PROJECT_NAME	Claude Code
--claude-session-uuid / HERALD_CLAUDE_SESSION_UUID	Claude Code
--non-interactive	Fail loud if a required value is missing — no prompts
--shell-target=zshrc bashrc profile	Override the shell-startup-file picker

Per Universal §11.4.10 the wizard never prints raw credentials back, masks values in the ~/.herald/credentials.md summary, and re-reads the shell-startup file to verify the write. Env-supplied values are never re-persisted (they're already in your shell file).

```
pherald wizard credentials telegram \
    --bot-token=80000000000:XXXX --chat-id=987654321 \
    --shell-target=zshrc --non-interactive
```

Detailed per-service walkthroughs: docs/guides/messengers/TELEGRAM.md, docs/guides/dispatchers/CLAUDE_CODE.md, docs/guides/OPERATOR_CREDENTIALS.md.

§9. commit-push + the §43 GitOps commands

pherald commit-push performs a §2-disciplined commit through a single locked endpoint (an O_CREATE|O_EXCL commit-lock under .git/) and, with --push, fans the commit out to every configured mirror remote. It wraps the canonical constitution commit_all.sh when discoverable.

Flag	Meaning
-m, --message string	Commit message (required)
--push	Fan the commit out to every configured mirror remote
--scope strings	Paths to stage (default: all tracked changes)
--repo string	Repo dir (default: discovered from CWD)
--emit	Also drive the §2 verdict as a real constitution event

The remaining §43 project-lifecycle GitOps commands surfaced by `pherald --help --fetch-guard` (§11.4.37 pre-edit fetch+rebase), `install-upstreams` (§11.4.36 mirror-remote config from `upstreams/*.sh`), `pre-push` (§11.4.71 fetch+investigate+integrate hook), `reopen` (§11.4.55 Issues→Fixed reversal), `submodule-propagate` (§3 owned-submodule walk) — are all wired (v1.0.0 Batch C, HRD-029/030/043/044/049/053). Run `pherald <command> --help` for each one's flag surface.

§10. Environment variable reference

Variable	Used by	Required?
HERALD_PG_DSN	serve, migrate	required for both
HERALD_AUTH_MODE	serve	required (hmac/jwks)
HERALD_AUTH_HMAC_SECRET	serve	required when HERALD_AUTH_MODE=hmac
HERALD_REDIS_URL	serve	optional (enables hot idempotency cache)
HERALD_TGRAM_BOT_TOKEN	serve (channel), listen, wizard	required for listen
HERALD_TGRAM_CHAT_ID	listen, wizard	required for listen
HERALD_PROJECT_NAME	listen (CC session name)	optional
HERALD_CLAUDE_BIN	listen, wizard	optional
HERALD_WORKABLE_DB	listen, watch	optional (DB path default)
HERALD_CHANNELS + per-channel env	watch, listen (fan-out)	per channel config

Resolution order project-wide: exported shell vars win; `.env` is fallback only.

§11. References

- Source: `pherald/cmd/pherald/main.go` (+ `serve.go`, `listen.go`, `watch.go`, `migrate.go`, `wizard*.go`, `gitops_cmds.go`, `stubs.go`).
- Branding: `commons/branding.go` (flavor p).
- Shared CLI scaffold: `commons/cli/` (`NewRootCmd`, `VersionCmd`).
- Integration walkthrough: `docs/INTEGRATION.md` (§5–§9 cover serve + first CloudEvent).
- Credentials: `docs/guides/OPERATOR_CREDENTIALS.md`, `docs/guides/messengers/TELEGRAM.md`, `docs/guides/dispatchers/CLAUDE_CODE.md`.
- Companion flavor guides: `docs/guides/{SHERALD, CHERALD, BHERALD, RHERALD, IHERALD, SCHERALD, QAHERALD}.md`.

Sources verified

Verified 2026-05-30: internal doc — no external online sources. Every subcommand, flag, default, env var, and "live vs not-yet-implemented" claim above was derived by **running the built pherald binary** (`pherald --help`; `pherald serve|listen|watch|migrate|wizard|wizard credentials|commit-push --help`; `pherald version --json`) and reading `pherald/cmd/pherald/main.go + commons/branding.go` on 2026-05-30. No flags were invented. Re-verify whenever the pherald Cobra surface changes (new subcommand, flag rename, or `migrate down/validate landing`).